

Technical Challenges

Hardware Limitations

The main issue during this project was the hardware. The computer used does not have a GPU (Graphics Processing Unit). Since the AI models used for transcription and speech synthesis are very heavy, I had to make several changes to get them to run on a standard CPU.

Long Processing Times

Because of the lack of a GPU, processing took a very long time. For example, a relatively short video about the "Strait of Hormuz disruption threatens to shake the global economy" took about 4 hours to complete. Because of these time constraints, I decided to use an even shorter video (1 minute and 26 seconds) for the final test. Even with this shorter video, the process still took nearly an hour to run.

UI and Browser Timeouts

When using the web interface (localhost:8501), the process would start normally, but the browser would eventually time out and show an "Error" during the speech synthesis (TTS) stage. By checking the terminal, I could see that the backend was actually still working and producing the correct .json and .wav files, but the "Stitch" section would never start because the UI had an Error status. Due to the Stitch segment of the pipeline never starting the parts that would compose a video were never stitched together, also the .vtt file that would contain the subtitles in spanish were never generated.

Manual Pipeline Execution

To overcome the UI crashes and timeout, trying to get the whole video in one execution, I moved the work into a Python notebook and used the command line to finish the job.

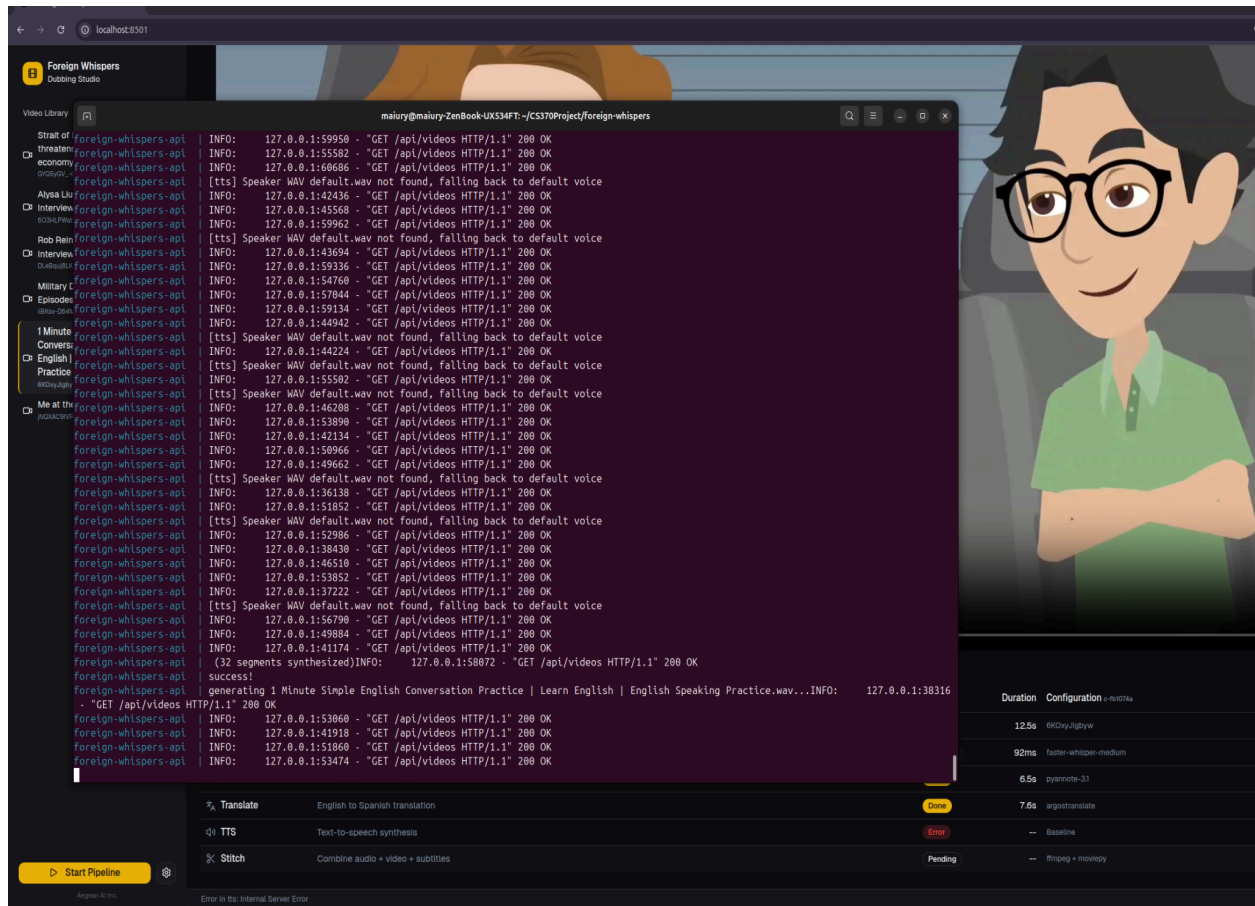
Since the automatic subtitle generation also failed due to the timeout, I had to:

- Manually use the API endpoint (localhost:8080) to generate the Spanish subtitle file.
- Use the ffmpeg command in the terminal to manually stitch the video, Spanish audio, and subtitles together into the final file.

While these manual steps were not the original plan for a "one-click" pipeline, they allowed me to successfully complete the project and produce a working dubbed video in Spanish despite the hardware limitations. I am aware of the synchronization and speech generation issues, but chose to focus on producing a complete dubbed video.

Project Links:

- Video on YouTube: <https://youtu.be/tlNWD75v6Do>
- GitHub Repository: <https://github.com/mdse789/CS370Project>



The picture shows the UI with an error message in the TTS section of the pipeline, while the terminal shows the TTS synthesis still happening.